Max **Simmonds, MEng, MIET**

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Profile

A driven, technically minded individual, always pushing to achieve. Confident and ambitious, I strive for perfection. Experience with working in both space and aerospace companies, from start-ups to the world's largest particle physics laboratory, interdisciplinary teams, highly technical positions, and customer facing roles; I have exceptional social, engineering, and presentation skills. Currently working towards chartered engineering status. Looking to develop my leadership skills, and push boundaries within business.

Achievements

- First Class Master's degree, with an aggregate mark of 84%
- Voted "Bright Spark 2019" by Electronics Weekly / RS Components
- Co-author of a paper on discrete modelling of switch mode converters
- Manufacture Top 100 (exemplary award) 2015 - a prestigious list of exceptional individuals in the UK
- Previously certified LabVIEW Associate Developer (CLAD) & Certified LabVIEW Developer (CLD)
- Presented at the Royal Institute of Science with Professor Danielle George (hosted by Professor Brian Cox)
- Two published articles in engineering journals - See LinkedIn profile for full details
- Developer of world first myRIO powered musical Tesla coil and self-balancing electric unicycle - See LinkedIn profile for full details

Relevant Experience

Thales Alenia Space
Bristol, England

TECHNICAL ELECTRONICS LEAD FOR CRYOGENIC COOLERS — *Oct 2021 - Present*

- After two months, I was promoted to the technical lead of the control electronics due to my motivation, technical and leadership skills
- I developed the prototype electronics for the cooler driving system, including firmware development on an STM32. I wrote register level drivers for the UART (for a user terminal), the PWM, PLL, and numerically controlled oscillator to generate sine wave outputs
- I wrote MATLAB scripts to run system identification of captured data, generating electrical equivalent models, and bode plots/
- I developed several novel and innovative designs, using model-based development. Including a new voltage-based control scheme, a passive launch locking system to limit piston excursions during launch, and representative electrical load modelling a cryogenic cooler. All reduced cost & complexity, along with space.
- Work package manager, involving hour tracking/estimating, following up with product orders, communicating with manufactures and stake holders. Managed several teams, including mechanical, systems, thermal, electronics, and quality.

The Engineering Octopus
Bristol, England

FOUNDER & DESIGN ENGINEER — *Apr 2020 - Present*

- Founded an electronic consultancy, focusing on electronic design
- Range of customers from the mining industry through to musicians, fuel cell manufacturing companies, and other well established design consultancies
- Successfully turning over £50k a year, working part time, alongside a full-time role
- Managing the marketing, business communication, networking, admin, and accounting

Diamond Light Source LTD
Oxfordshire, England

POWER SUPPLIES ENGINEER — *Sep 2020 - Oct 2021*

- Lead & performed a critical design review of complex, multi-layer, multi-signal board with an on-board FPGA for digital control of a power supply, specifically targeting the analogue front end, solving several issues related to error induced noise & premature component failures
- Implemented a new engineering change request process
- Design, schematic capture, & layout of high precision (5ppm) current transducer for digital power supply, achieving 1uA precision and decreasing drift levels by a factor of 2
- Developed test plans & procedures
- Received the highest rating in Personal Development Review of 'exceptional'

Open Cosmos
Oxfordshire, England

Electronic Engineer - Electrical Power Subsystem & Backplane
Jul 2019 - Sep 2020

- Design, schematic, layout, & testing of 4 layer, digitally controlled power conditioning & distribution unit (PCDU) for CubeSats
- Development & test of new designs for the electrical power subsystem (solar arrays, analogue/digital maximum power point tracking (MPPT) circuits, photovoltaic side converters, battery charge regulators, cell balancing, & power distribution/conditioning
- Developed digital controller for DCDC converter, implementing as an IIR filter on MCU
- Increased tracking accuracy of MPPT by 15%, & increased power density
- Developing models and simulations of hardware, for increase speed of testing & development times
- Working as part of a fast-paced, agile team, developing CubeSats to make Space accessible
- Fully solving several critical issues, under pressure, quickly & effectively
- Writing test scripts for automated testing of subsystems - including stress testing
- Writing requirements, test procedures, board bring up tests, & verification tests

Safran (UK) Electrical & Power
Pitstone, England

Senior Specialist Engineer - Hardware Design (Research & Technology)
Aug 2018 - Jul 2019

- Designing a new generator control unit (GCU) for more electric aircraft, from system requirements through to production
- Leading trade-off for next generation power supplies - comparing flyback, two-switch forward, & phase shifted bridge converter
- Design of the power system architecture, analogue circuit design, reverse engineering, digital implementation of analogue circuits, modelling & simulation of designs
- Design, schematic capture, & review of PCB for exciter field drive to 140kVA generators
- Initial testing of digital control unit on real generators in test cells
- Writing test procedures, board bring up tests, & verification tests

European Space Agency (ESA)
Noordwyk, Netherlands

Power Systems Engineer (Young Graduate Trainee)
July 2017 - July 2018

- A research based project testing the feasibility of digital control (via an FPGA) for bidirectional DC/DC converters in Space applications, resulting in co-authoring of a paper based on my work - link to paper
- Developed a 100W Buck + Boost, meeting all design requirements (ECSS) & having a very small footprint. Less than 0.5% static regulation, & 1% dynamic regulation
- Developed mathematical models of the converter & digital controller, simulated to test requirements, & programmed in VHDL. Further simulation was done using ModelSim. PCB layout (minimising current loops, ESR, EMC, etc.) and manufacturing using ESA's power lab. Fully tested, checking every aspect of the converter, before optimising for the final design
- Attended weekly engineering meetings, offering knowledge where possible. Attended a week long training for electric propulsion systems, & several concurrent design facility meetings to help develop satellites for the Space science
- **Required:** Deep knowledge of power electronic converters, power semiconductor operation, ECSS requirements & converter typologies

CERN - European Organisation for Nuclear Research
Geneva, Switzerland

Application Engineering Summer Student
July 2016 - August 2016

- 2-month summer placement working full time at CERN & undergoing lectures to further knowledge in big Physics and electronics
- Project to reverse engineer, upgrade, & restore existing fibre optic notch filter - used in the Antiproton Decelerator in CERN
- The project was completed, tested, & documented. Now published on the CERN public document server
- **Required:** Developing large LabVIEW applications, physics, electronic & mechanical, control engineering, & RF system design/test

National Instruments (NI)
Newbury, England

Application Engineer & Technical Marketing Engineer (Industrial Placement)
July 2014 - August 2015

- Achieved two of three possible LabVIEW certifications. 'Certified LabVIEW Developer' was the highest
- Developer of world first myRIO powered musical Tesla coil - https://youtu.be/AyXX_V5bcWM
- Solved customer issues regarding NI products. Receiving an average of 92% customer satisfaction rating
- Presented at technical conferences & taught customer education classes

Skills

Programming & Engineering
- Programming Languages - VHDL, C, Embedded C, C++, Python, LabVIEW, MATLAB, Arduino, PHP, HTML, JavaScript, TwinCat2, Assembly, LaTeX
- PCB Design - Altium Designer, KiCAD, Eagle, Cadence Allegro/CIS, NI Ultiboard, Proteus
- Simulation - SIMetrix, Simulink, LTspice, Proteus, ModelSim
- Mathematics Software Packages - MATLAB, Maple, WolframAlpha, Scilab
- Knowledge of git and source code control
- Knowledge of Linux and Windows operating systems

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Volunteering & Leadership

- Project manager (MEng project) for a central government agency
- LabVIEW Student Ambassador (organising LabVIEW training)
- Outreach Ambassador at Plymouth University for STEM subjects
- Completed first stage of the Engineering Leadership Program (ELP) at National Instruments

Education

University of Plymouth *Plymouth, England*

MEng Electrical and Electronic Engineering, 5yr with placement, (**First Class Degree**) *2012 - 2017*

- Graduated with a first-class Master's degree, making the Dean's Hour Role list every year
- Project manager for confidential MEng project (Inertial Navigation System in GPS restricted environments) for a government agency. Achieved first class grade, involved mathematical modelling of the Earth's rotation through different frames of references, novel probabilistic models, and algorithms
- Developed an electric self-balancing unicycle for BEng dissertation. Used BLDC hub motor, involved designing of 1000W 3 phase inverter. Required a good understanding of control theory & sensor fusion - winning 3 awards. See LinkedIn profile for full details
- Built and tested a grid-tied inverter, developing a recursive DFT algorithm for quick phase locking, implemented down to bit-level
- Built and tested a 3 element Yagi antenna for listening to the International Space Station
- **Modules** - Microcontrollers & Embedded programming, Digital & Analogue Electronics, Classical & Modern Control Theory, RF Circuits & System Design, Wireless Communications, DSP, Renewable Energy, Advanced Power Systems, Nanotechnology & Nanoelectronics

Honours & Awards

Bright Sparks 2019 *London, England*

Electronics Weekly / RS Components *June 2019*

Electronics Weekly and RS Components selected me as one of the UK's brightest and talented young electronic engineers

Engineering Excellence, Technical Innovation, Contribution to Student Life *Plymouth, England*

Plymouth University *May 2016*

'Contribution to student life' awarded by my peers, voting for someone who has helped them during their studies, been kind/supportive. 'Engineering excellence' & 'Technical innovation' awarded for my self-balancing, electric unicycle

The Manufacture Top 100 (Exemplary Award) *Birmingham, England*

The Manufacturer *October 2015*

The Top 100 comprises of a report & event recognising an elite group of individuals identified as manufacturing champions. Out of the winners, twenty people are profiled in depth, to underline their outstanding contribution to the sector and marked as exemplary

Dean's Honour Role List *Plymouth, England*

Plymouth University *2012-2017*

Students who achieve an aggregate mark of over 70% in pre-final stages of their undergraduate degree programme or achieve a first-class degree in the final year classification qualify for the list

UK Winners & Top 5 Globally *Met Office, England*

NASA Space Apps Challenge *February 2013*

A 48-hour Hackathon developing (in a team) a nanosatellite that integrated a camera, image tracking algorithm, and a RaspberryPi. It was exhibited at the Victoria and Albert Museum in London, and the team received an honourable mention on NASA's homepage.

Certifications

- Certified LabVIEW Associate Developer (CLAD) and Certified LabVIEW Developer (CLD)
- NVQ Level 2 in Engineering

Extracurricular Activity

Latching Current Limiter

- Design, schematic, layout, and manufacture of a fully analogue latching current limiter (digital/e-fuse) to increase understanding of analogue electronics

Designed and Developed A 3D Printer

- Built a fully functional 3D printer (my own design) from wood/metal, printed 5 generations of itself, each more precise than the last

Designed a Control box & Power System for 7.5KW Powder Coating Oven

- A system that used a PID controller to reach a certain temperature, with functions such as pause, start, stop, over temperature alarm, & timer done. Temperature stabilisation done using a SSR and PID controller.

Interests

Rock climbing, Motocross, Electric Vehicles & Renewable Energy Technology, Motorcycles, Classic Cars, Electronics, Hiking, and Space.